

There's a new sheriff in Town

PCs have been associated with the x86 architecture for the longest time. Apple made a dent in the x86 market share by dropping Intel in favour of their own ARM-based processors about three years ago. Apple being a stickler for user experience had its own software ecosystem prepared for this move and software vendors whose products were popular on Macs also came out with native software. And now ARM makes up a good 12.8 per cent of the PC market. But Apple's pricey laptops and positioning as a premium brand keeps it from ever achieving the lion's share. It seems that's all going to change, thanks to Qualcomm.

Its announcement of the Snapdragon X Elite SoC is an emerging disruption, heralding an intriguing shift in the microprocessor ecosystem. The Snapdragon X Elite, with its new Oryon CPU cores, is a pivotal leap that symbolises Qualcomm's sisphean endeavour to engrave its footprint in the PC market. The new ARM-based SoC not only breaks the mold but signifies an earnest endeavor to challenge the hegemony of Intel and AMD in the notebook sector.

With the Snapdragon X Elite, Qualcomm is not merely tossing a pebble into the pond; it's hurling a boulder. The Snapdragon X Elite's cornerstone is its Oryon CPU cores, a product of its acquisition of Nuvia Labs in 2021. This acquisition was a strategic move to amass a reservoir of intellectual prowess from industry veterans, notably Gerard Williams, a former Chief CPU Architect at Apple. The person who has worked at ARM for about 12 years before spending a further 9 years at Apple heading the team that built all the SoCs from the Apple A7 through the A12. You can see why the Oryon cores represent a paradigm shift.

This shift is reflective of a broader industry trend where microprocessor designs are continuously evolving to strike a balance between power efficiency and performance. The preliminary benchmarks from Qualcomm, although yet to be independently verified, suggest a significant margin of performance improvement over Apple's M2 Max in single-threaded benchmarks, and a promising competitive stance against Intel's Core i9-13980HX, albeit at 70 per cent less power.



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These claims, if substantiated, could recalibrate the dynamics of the x86 industry. Moreover, the Snapdragon X Elite's comprehensive I/O portfolio, coupled with the Adreno GPU and Hexagon Neural Processing Unit (NPU), underscores a concerted effort to cater to a broader spectrum of computing demands, from AI workloads to high-fidelity graphics. The Hexagon NPU's initial benchmarks already put it at 10x of the competition while the Adreno integrated GPU is a sliver ahead if not at par with the competition.

However, the Snapdragon X Elite's foray into the realms traditionally governed by x86 architectures is not devoid of hurdles. The allegiance of major OEMs like Lenovo, HP, and Microsoft is encouraging, yet the real litmus test will be the widespread adoption and the ecosystem support, particularly from software developers. The SoC's compatibility with prevalent instruction sets and its ability to play well with discrete GPUs from NVIDIA and AMD, especially in the burgeoning gaming laptop segment, will definitely help build its image as a jack-of-all-trades solution. Being a little sceptical of not having seen other major OEMs aside from HP and Leonovo at the announcement, I posed a simple question to Alex Katouzian, SVP & GM of the Mobile, Compute and XR divisions about how many ODMs had actually signed up to create laptop designs based on Snapdragon X Elite. And before I could even complete my question, he butted in with a confident, "All of them". I was expecting a response more on the lines of "We're still in talks" but his response had me thoroughly intrigued. Moreover, the tentative mid-2024 launch window of Snapdragon X Elite has set the stage for a riveting competition.

The Snapdragon X Elite, with its blend of innovative CPU design and a bouquet of enticing features, is Qualcomm's gambit to penetrate the Windows-on-Arm laptop market. The resonance of this venture could potentially redraw the contours of the notebook processor landscape. As we inch closer to a new epoch, the dynamics between Intel, AMD, and emergent players like Qualcomm will be riveting to observe. The traditional bastions of x86 processors are being challenged, and the reverberations will be felt across the processor manufacturing landscape. **d**